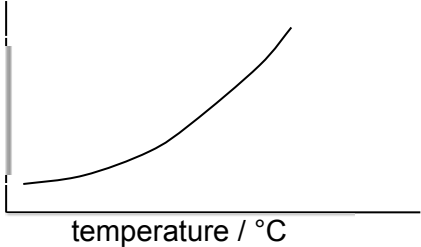


1(a)(i)	<i>independent variable</i> : temperature ; <i>dependent variable</i> : distance travelled by liquid / volume of oxygen taken up ;	2
1(a)(ii)	1 (because) these are temperatures at which, enzymes are expected to be functioning / woodlice may survive ; 2 (intervals of) 5 °C / 3 °C / 2 °C ;	1
1(a)(iii)	inert material / glass beads, of same volume (as woodlice) ; A dead (sterilised) woodlice	1
1(a)(iv)	<i>idea that</i> absorption of carbon dioxide allows measurement of uptake in oxygen / AW ;	1
1(a)(v)	<i>any 1 from</i> : 1 movement of woodlice ; 2 size / age / sex, of woodlice ; 3 AVP ;	1
1(b)(i)	<i>allow in context of using the same respirometer for each temperature, or separate respirometers set up in the same way for each temperature</i> <i>any 7 of</i> : 1 <i>idea of</i> adding a fixed volume of carbon dioxide absorbent solution ; 2 <i>idea of</i> adding fixed, number / mass, of woodlice to tube ; 3 <i>idea of</i> method of adding dye solution to capillary tube ; 4 <i>idea of</i> attaching tubes and making airtight ; 5 <i>idea of</i> placing respirometer tube into a water-bath at a stated temperature ; 6 <i>idea of</i> leaving apparatus to equilibrate with tap open ; 7 <i>idea of</i> closing the tap and, marking / recording, position of liquid ; 8 <i>idea of</i> measuring movement of dye on the scale for, 5 minutes / a set time ; 9 <i>idea of</i> re-opening tap / using syringe, to allow liquid to equalise and then closing to take another measurement ; 10 <i>idea of</i> a minimum of three measurements (in sequence) <u>and</u> taking a mean ; 11 <i>idea of</i> repeating experiment at each temperature, e.g. 5 °C, 10 °C, 15 °C, etc ; 12 low risk experiment or medium risk experiment due to (named) carbon dioxide absorbent is harmful <u>and</u> wear gloves ; 13 AVP ;	7

Question	Answer	Marks
1(b)(ii)	1 <i>idea of</i> converting the distance in cm to mm³ by multiplying by 10 ; 2 volume calculated = volume of oxygen absorbed by woodlice ; 3 divide each of the volumes by time (oxygen uptake per unit time = rate of respiration) ;	3
1(b)(iii)	1 axes correctly orientated with labels ; 2 axes have units ; 3 line shows (exponential) increase as temperature increases ; e.g. 	3
1(c)	woodlice are using mostly carbohydrate as their energy source ; A using a mixed energy source A using, protein / lipid, under anaerobic conditions R using, protein / fat only	1